

# Complex Analysis Homework 1

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**7.** The family of mappings introduced here plays an important role in complex analysis. These mappings, sometimes called Blaschke factors, will reappear in various applications in later chapters.

(a) Let  $z, w$  be two complex numbers such that  $\bar{z}w \neq 1$ . Prove that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| < 1 \quad \text{if } |z| < 1 \text{ and } |w| < 1,$$

and also that

$$\left| \frac{w - z}{1 - \bar{w}z} \right| = 1 \quad \text{if } |z| = 1 \text{ or } |w| = 1.$$

[Hint: Why can one assume that  $z$  is real? It then suffices to prove that

$$(r - w)(r - \bar{w}) \leq (1 - rw)(1 - r\bar{w})$$

with equality for appropriate  $r$  and  $|w|$ .]

(b) Prove that for a fixed  $w$  in the unit disc  $\mathbb{D}$ , the mapping

$$F : z \mapsto \frac{w - z}{1 - \bar{w}z}$$

satisfies the following conditions:

- (i)  $F$  maps the unit disc to itself (that is,  $F : \mathbb{D} \rightarrow \mathbb{D}$ ), and is holomorphic.
- (ii)  $F$  interchanges 0 and  $w$ , namely  $F(0) = w$  and  $F(w) = 0$ .
- (iii)  $|F(z)| = 1$  if  $|z| = 1$ .
- (iv)  $F : \mathbb{D} \rightarrow \mathbb{D}$  is bijective. [Hint: Calculate  $F \circ F$ .]

**证明** (a) 考虑  $|\frac{w-z}{1-\bar{w}z}|^2 = \frac{(w-z)(\bar{w}-\bar{z})}{(1-\bar{w}z)(1-w\bar{z})}$

故只需要证明

$$(w-z)(\bar{w}-\bar{z}) < (1-\bar{w}z)(1-w\bar{z})$$

化简得

$$w\bar{w} + z\bar{z} < 1 + w\bar{w}z\bar{z}$$

, 即

$$(1-|w|^2)(1-|z|^2) > 0$$

根据  $|z| < 1$  and  $|w| < 1$ ,  $(1-|w|^2)(1-|z|^2) > 0$  成立

第二个式子同理

(b)

(i) 由 a 即可得 " $F$  maps the unit disc to itself", 由于  $w-z$  和  $1-\bar{w}z$  都是纯的且  $1-\bar{w}z \neq 0$ , 于是  $F$  是纯的

(ii) 代入计算即得

(iii) 根据 a 即得

(iv) 在  $\mathbb{D}$  上计算有  $F \circ F = id$ , 因此  $F$  是双射

**8.** Suppose  $U$  and  $V$  are open sets in the complex plane. Prove that if  $f : U \rightarrow V$  and  $g : V \rightarrow \mathbb{C}$  are two functions that are differentiable (in the real sense, that is, as functions of the two real variables  $x$  and  $y$ ), and  $h = g \circ f$ , then

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z}$$

and

$$\frac{\partial h}{\partial \bar{z}} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial \bar{z}} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial \bar{z}}.$$

This is the complex version of the chain rule.

**证明** 设  $f = u + iv, g = g(v, u)$ , 从而

$$\frac{\partial h}{\partial x} = \frac{\partial g}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial x}$$

$$\frac{\partial h}{\partial y} = \frac{\partial g}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial y}$$

于是

$$\begin{aligned}
\frac{\partial h}{\partial z} &= \frac{1}{2} \left( \frac{\partial h}{\partial x} + \frac{1}{i} \frac{\partial h}{\partial y} \right) \\
&= \frac{1}{2} \left( \frac{\partial g}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial x} + \frac{1}{i} \left( \frac{\partial g}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial y} \right) \right) \\
&= \frac{1}{2} \left( \frac{\partial g}{\partial u} \left( \frac{\partial u}{\partial x} + \frac{1}{i} \frac{\partial u}{\partial y} \right) + \frac{\partial g}{\partial v} \left( \frac{\partial v}{\partial x} + \frac{1}{i} \frac{\partial v}{\partial y} \right) \right) \\
&= \frac{\partial g}{\partial u} \frac{\partial u}{\partial z} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial z}
\end{aligned} \tag{1}$$

根据

$$\frac{\partial f}{\partial z} = \frac{\partial u}{\partial z} + i \frac{\partial v}{\partial z}, \quad \frac{\partial \bar{f}}{\partial z} = \frac{\partial u}{\partial z} - i \frac{\partial v}{\partial z}$$

, 可得

$$\frac{\partial u}{\partial z} = \frac{1}{2} \left( \frac{\partial f}{\partial z} + \frac{\partial \bar{f}}{\partial z} \right), \quad \frac{\partial v}{\partial z} = -\frac{i}{2} \left( \frac{\partial f}{\partial z} - \frac{\partial \bar{f}}{\partial z} \right) \tag{2}$$

而根据

$$\frac{\partial g}{\partial z} = \frac{1}{2} \left( \frac{\partial g}{\partial u} + \frac{1}{i} \frac{\partial g}{\partial v} \right), \quad \frac{\partial g}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial g}{\partial u} - \frac{1}{i} \frac{\partial g}{\partial v} \right)$$

有

$$\frac{\partial g}{\partial u} = \frac{\partial g}{\partial z} + \frac{\partial g}{\partial \bar{z}}, \quad \frac{\partial g}{\partial v} = i \left( \frac{\partial g}{\partial z} - \frac{\partial g}{\partial \bar{z}} \right) \tag{3}$$

将 (2)(3) 代入 (1), 即有

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z}$$

第二个等式同理可得

**9.** Show that in polar coordinates, the Cauchy-Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\log z = \log r + i\theta \quad \text{where } z = re^{i\theta} \text{ with } -\pi < \theta < \pi$$

is holomorphic in the region  $r > 0$  and  $-\pi < \theta < \pi$ .

**证明** 根据  $x = r \cos \theta, y = r \sin \theta$  和链式法则, 有

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{1}{r} \sin \theta, \quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \sin \theta + \frac{\partial u}{\partial \theta} \frac{1}{r} \cos \theta$$

二者联立解得

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta, \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial u}{\partial x} \sin \theta + \frac{\partial u}{\partial y} \cos \theta$$

类似的, 可以得到

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x} \cos \theta + \frac{\partial v}{\partial y} \sin \theta, \frac{1}{r} \frac{\partial v}{\partial \theta} = -\frac{\partial v}{\partial x} \sin \theta + \frac{\partial v}{\partial y} \cos \theta$$

结合柯西-黎曼方程  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ , 即有

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

对于  $\log z = \log r + i\theta$ ,  $u = \log r, v = \theta$ , 于是在定义域上  $y, v$  处处可微分, 且满足

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} = \frac{1}{r} \quad \text{and} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r} = 0$$

于是  $\log z$  是纯的

10. Show that

$$4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta,$$

where  $\Delta$  is the Laplacian

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

证明 已知:

$$\frac{\partial}{\partial z} = \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

于是

$$\begin{aligned} 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} &= 4 \cdot \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \cdot \frac{1}{2} \left( \frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \\ &= \left( \frac{\partial}{\partial x} \right)^2 - \left( i \frac{\partial}{\partial y} \right)^2 \\ &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \Delta \end{aligned}$$

同理可证:

$$4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta.$$

16. Determine the radius of convergence of the series  $\sum_{n=1}^{\infty} a_n z^n$  when:

(a)  $a_n = (\log n)^2$

(b)  $a_n = n!$

(c)  $a_n = \frac{n^2}{4^n + 3n}$

(d)  $a_n = \frac{(n!)^3}{(3n)!}$  [Hint: Use Stirling's formula, which says that  $n! \sim cn^{n+\frac{1}{2}}e^{-n}$  for some  $c > 0$ .]

(e) Find the radius of convergence of the hypergeometric series

$$F(\alpha, \beta, \gamma; z) = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)\beta(\beta+1) \cdots (\beta+n-1)}{n!\gamma(\gamma+1) \cdots (\gamma+n-1)} z^n.$$

Here  $\alpha, \beta \in \mathbb{C}$  and  $\gamma \neq 0, -1, -2, \dots$

证明 (a)

$$|a_n|^{1/n} = ((\log n)^{1/n})^2$$

令  $t = \log n$ , 则  $n = e^t$ , 从而  $(\log n)^{1/n} = t^{1/e^t} = e^{\frac{\log t}{e^t}}$ , 当  $n \rightarrow \infty$  时  $t \rightarrow \infty$ ,  $\lim_{t \rightarrow \infty} \frac{\log t}{e^t} = 0$ , 故  $\lim_{n \rightarrow \infty} (\log n)^{1/n} = e^0 = 1$ 。因此:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = 1^2 = 1$$

收敛半径:  $R = \frac{1}{1} = 1$ 。

(b) 利用斯特林公式  $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$  :

$$|a_n|^{1/n} = (n!)^{1/n} \sim \left(\sqrt{2\pi n} \left(\frac{n}{e}\right)^n\right)^{1/n} = (2\pi n)^{1/(2n)} \cdot \frac{n}{e}$$

由于  $\lim_{n \rightarrow \infty} (2\pi n)^{1/(2n)} = 1$ , 故:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \frac{n}{e} = \infty$$

因此:

$$R = 0$$

(c)

$$|a_n|^{1/n} = \left(\frac{n^2}{4^n + 3n}\right)^{1/n} = \frac{(n^{1/n})^2}{(4^n (1 + \frac{3n}{4^n}))^{1/n}} = \frac{(n^{1/n})^2}{4 \cdot (1 + \frac{3n}{4^n})^{1/n}}$$

而  $\lim_{n \rightarrow \infty} n^{1/n} = 1$ , 且  $\lim_{n \rightarrow \infty} \frac{3n}{4^n} = 0$ , 故  $\lim_{n \rightarrow \infty} \left(1 + \frac{3n}{4^n}\right)^{1/n} = 1$ 。

因此:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{4 \cdot 1} = \frac{1}{4}$$

收敛半径:  $R = \frac{1}{1/4} = 4$ 。

(d) 由斯特林公式:

$$(n!)^3 \sim (2\pi n)^{3/2} \left(\frac{n}{e}\right)^{3n}, \quad (3n)! \sim \sqrt{6\pi n} \left(\frac{3n}{e}\right)^{3n}$$

代入

$$|a_n|^{1/n} \sim \left(\frac{(2\pi n)^{3/2} \left(\frac{n}{e}\right)^{3n}}{\sqrt{6\pi n} \left(\frac{3n}{e}\right)^{3n}}\right)^{1/n} = \left(\frac{(2\pi)^{3/2}}{\sqrt{6\pi}}\right)^{1/n} n^{1/n} \cdot \frac{1}{27}$$

$\lim_{n \rightarrow \infty} (n)^{1/n} = 1$ , 故:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = 1 \cdot \frac{1}{27} = \frac{1}{27}$$

收敛半径:  $R = \frac{1}{1/27} = 27$ 。

(e) 超几何级数通项系数:

$$a_n = \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)\beta(\beta+1) \cdots (\beta+n-1)}{n!\gamma(\gamma+1) \cdots (\gamma+n-1)}$$

若  $-\alpha \in \mathbb{N}$  或  $-\beta \in \mathbb{N}$ , 则从某一项起  $a_n = 0$ , 从而收敛半径为  $R = \infty$

若  $-\alpha \notin \mathbb{N}$  且  $-\beta \notin \mathbb{N}$  时, 有

$$\frac{a_{n+1}}{a_n} = \frac{(\alpha+n)(\beta+n)}{(n+1)(\gamma+n)} \rightarrow 1 \quad (n \rightarrow \infty)$$

因此:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = 1$$

收敛半径:  $R = 1$

**23.** Consider the function  $f$  defined on  $\mathbb{R}$  by

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ e^{-1/x^2} & \text{if } x > 0. \end{cases}$$

Prove that  $f$  is indefinitely differentiable on  $\mathbb{R}$ , and that  $f^{(n)}(0) = 0$  for all  $n \geq 1$ . Conclude that  $f$  does not have a converging power series expansion  $\sum_{n=0}^{\infty} a_n x^n$  for  $x$  near the origin.

**证明** 当  $x > 0$  时,  $f(x) = e^{-1/x^2}$  是初等函数的复合, 显然无穷次可微, 且其各阶导数均为形如  $P(1/x)e^{-1/x^2}$  的形式 ( $P(t)$  为关于  $t$  的多项式);

当  $x < 0$  时,  $f(x) = 0$ , 显然无穷次可微, 且各阶导数均为 0;

当  $x = 0$  时,  $n = 1$  时:

$$f'(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{e^{-1/x^2}}{x} = 0$$

假设对  $k \geq 1$ , 有  $f^{(k)}(0) = 0$ , 且当  $x > 0$  时,  $f^{(k)}(x) = P_k(1/x)e^{-1/x^2}$  ( $P_k(t)$  为多项式)。

$$f^{(k+1)}(0) = \lim_{x \rightarrow 0^+} \frac{f^{(k)}(x) - f^{(k)}(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{P_k(1/x)e^{-1/x^2}}{x} = 0$$

由归纳法, 对所有  $n \geq 1$ ,  $f^{(n)}(0) = 0$ 。

综上,  $f$  is indefinitely differentiable on  $\mathbb{R}$

若  $f(x)$  在原点附近有收敛的幂级数展开:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

由  $f^{(n)}(0) = 0$  ( $n \geq 1$ ), 得  $f(x) = f(0) = 0$ , 与  $x > 0$  时  $f(x) = e^{-1/x^2} > 0$  矛盾。

因此,  $f(x)$  在原点附近不存在收敛的幂级数展开。

**25.** The next three calculations provide some insight into Cauchy's theorem, which we treat in the next chapter.

(a) Evaluate the integrals

$$\int_{\gamma} z^n dz$$

for all integers  $n$ . Here  $\gamma$  is any circle centered at the origin with the positive (counterclockwise) orientation.

(b) Same question as before, but with  $\gamma$  any circle not containing the origin.

(c) Show that if  $|a| < r < |b|$ , then

$$\int_{\gamma} \frac{1}{(z-a)(z-b)} dz = \frac{2\pi i}{a-b},$$

where  $\gamma$  denotes the circle centered at the origin, of radius  $r$ , with the positive orientation.

**证明** (a) 设  $\gamma$  的参数方程为  $z = re^{i\theta}$  ( $\theta \in [0, 2\pi]$ ,  $r > 0$ ), 则  $dz = ire^{i\theta}d\theta$ , 代入积分得:

$$\int_{\gamma} z^n dz = \int_0^{2\pi} (re^{i\theta})^n \cdot ire^{i\theta} d\theta = ir^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta$$

当  $n \neq -1$  时:

$$\int_0^{2\pi} e^{i(n+1)\theta} d\theta = \frac{1}{i(n+1)} e^{i(n+1)\theta} \Big|_0^{2\pi} = 0$$

故  $\int_{\gamma} z^n dz = 0$ ;

当  $n = -1$  时:

$$\int_{\gamma} z^{-1} dz = ir^0 \int_0^{2\pi} e^{i0 \cdot \theta} d\theta = i \int_0^{2\pi} 1 d\theta = 2\pi i$$

综上:

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i & n = -1, \\ 0 & n \in \mathbb{Z}, n \neq -1. \end{cases}$$

(b)

参数化  $\gamma$ :  $z(\theta) = z_0 + Re^{i\theta}$ ,  $\theta \in [0, 2\pi]$ , 则  $dz = iRe^{i\theta}d\theta$ 。

积分化为:

$$\int_{\gamma} z^n dz = \int_0^{2\pi} (z_0 + Re^{i\theta})^n \cdot iRe^{i\theta} d\theta$$

当  $n \neq -1$  时:

$$\int_0^{2\pi} (z_0 + Re^{i\theta})^n \cdot iRe^{i\theta} d\theta = \frac{1}{n+1} [(z_0 + R)^{n+1} - (z_0 + R)^{n+1}] = 0$$

当  $n = -1$  时:

$$\int_{\gamma} z^n dz = \int_0^{2\pi} \frac{iRe^{i\theta}}{z_0 + Re^{i\theta}} d\theta = \ln(z_0 + R) - \ln(z_0 + R) = 0$$

综上, 对所有整数  $n$ , 均有:

$$\int_{\gamma} z^n dz = 0$$

(c)

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left( \frac{1}{z-a} - \frac{1}{z-b} \right)$$

$$\int_{\gamma} \frac{1}{z-a} dz = \int_0^{2\pi} \frac{1}{re^{i\theta}-a} \cdot ire^{i\theta} d\theta = i \int_0^{2\pi} \frac{re^{i\theta}}{re^{i\theta}-a} d\theta = i \int_0^{2\pi} \frac{1}{1-\frac{a}{r}e^{-i\theta}} d\theta.$$

由于  $|a| < r$ , 有  $|\frac{a}{r}e^{-i\theta}| < 1$ , 可将被积函数展开为几何级数:

$$\frac{1}{1-\frac{a}{r}e^{-i\theta}} = \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n e^{-in\theta},$$

逐项积分得:

$$i \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n \int_0^{2\pi} e^{-in\theta} d\theta.$$

而

$$\int_0^{2\pi} e^{-in\theta} d\theta = \begin{cases} 2\pi, & n=0, \\ 0, & n \neq 0. \end{cases}$$

故

$$\int_{\gamma} \frac{1}{z-a} dz = 2\pi i.$$

由 (b) 的结论

$$\int_{\gamma} \frac{1}{z-b} dz = \int_{\gamma+b} \frac{1}{z} dz = 0$$

综上

$$\int_{\gamma} \frac{1}{(z-a)(z-b)} dz = \frac{1}{a-b} \left( \int_{\gamma} \frac{1}{z-a} dz - \int_{\gamma} \frac{1}{z-b} dz \right) = \frac{2\pi i}{a-b}$$